

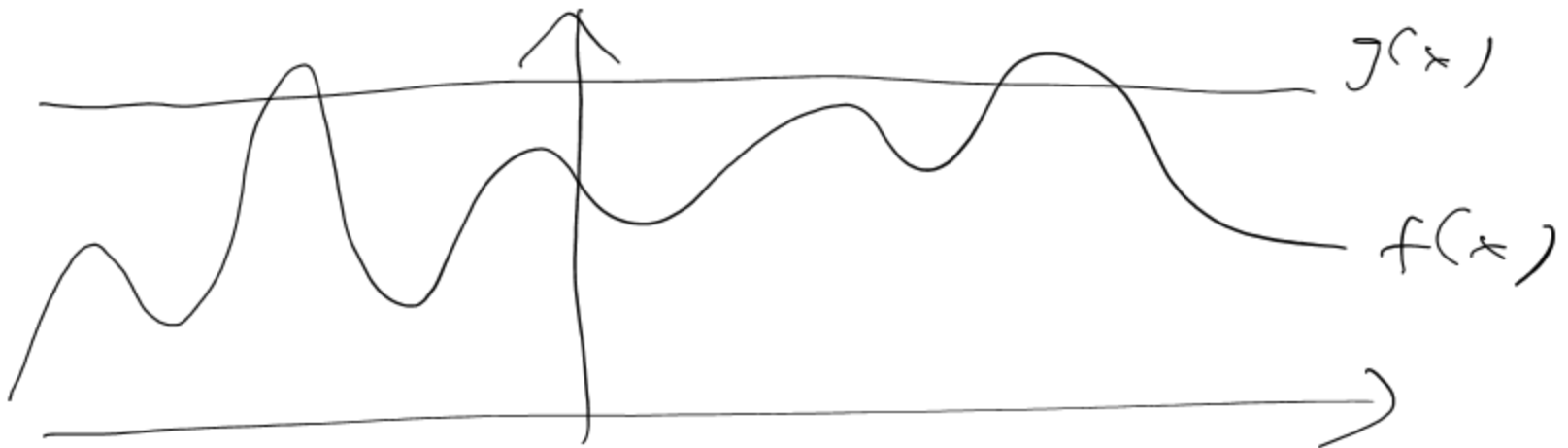
IEOR 4404: Simulation

Lecture 4: Acceptance-Rejection Method

Acceptance Rejection Method

A method to generate “difficult” r.v. from “easy” ones.

- We want to generate a r.v. with density $f(x)$ (difficult)
- We can generate a r.v. with density $g(x)$ (easy)



Acceptance Rejection Method

Acceptance rejection method:

- Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$  *presumably easy to do
e.g. use ITM*
 - “Accept” Y with probability $\frac{f(Y)}{Cg(Y)}$; otherwise “reject” and go back to the beginning
- The r.v. from $g(x)$ is called the proposal r.v.

Acceptance Rejection Method

Acceptance rejection method:

- Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$
 - Generate $U \sim Unif(0,1)$
 - If $U \leq \frac{f(Y)}{Cg(Y)}$, set $X = Y$ (accept); otherwise, go back to the beginning (reject)
 - Output X

Example

- Generate a r.v. with pdf

$$f(x) = 3x^2, \quad 0 \leq x \leq 1$$

- Use a *Unif*(0,1) proposal

$$g(x) = 1, \quad 0 \leq x \leq 1$$

A-R:

Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for } 0 \leq x \leq 1$$

$$\Rightarrow \frac{3x^2}{1} \leq C$$
$$\Rightarrow 3x^2 \leq C$$

↑
possible values taken by
the target r.v.

We can take $C = 3$

or $C = 100 \dots$

Procedure:

① Generate $Y \sim g(y) = \text{Unif}(0, 1)$

② "Accept" Y with prob $\frac{f(Y)}{Cg(Y)} = \frac{3Y^2}{3 \cdot 1}$
 $= Y^2$

otherwise "reject" and go back to ①

Pseudo-code:

- ① Generate $Y \sim \text{Unif}(0, 1)$
- ② Generate $U \sim \text{Unif}(0, 1)$
(indep of Y)
- ③ If $U \leq \frac{f(Y)}{c g(Y)} = Y^2$, then set $X = Y$
otherwise go back to ①

Output X

Example

- Generate a r.v. with pdf

$$f(x) = 3x^2, \quad 0 \leq x \leq 1$$

- Use a *Unif*(0,1) proposal

$$g(x) = 1, \quad 0 \leq x \leq 1$$

- We have

$$\frac{f(x)}{g(x)} = \frac{3x^2}{1} = 3x^2 \leq 3 \text{ for any } 0 \leq x \leq 1$$

- Thus we can choose $C = 3$

Example

Acceptance-rejection method:

- Generate $Y \sim Unif(0,1)$
- Generate $U \sim Unif(0,1)$
- If $U \leq \frac{f(Y)}{cg(Y)} = Y^2$, set $X = Y$ (accept); otherwise, go back to the beginning (reject)
- Output X

Example

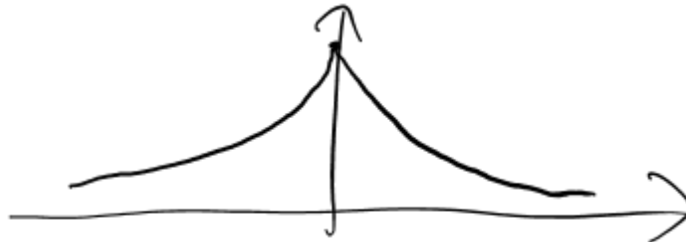
- Generate the standard normal r.v.

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

- Use a proposal

$$g(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

two-sided exponential pdf



Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for} \quad -\infty < x < \infty$$

$$\therefore \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}{\frac{1}{2} e^{-|x|}} = \sqrt{\frac{2}{\pi}} e^{-\frac{x^2}{2} + |x|}$$

Take

$$C = \max_{-\infty < x < \infty} \int \frac{2}{\pi} e^{-\frac{x^2}{2}} f(x)$$

$$= \int \frac{2}{\pi} e^{\max_{-\infty < x < \infty} \left\{ -\frac{x^2}{2} + |x| \right\}}$$

$$= \int \frac{2}{\pi} e^{\max_{x \geq 0} \left\{ -\frac{x^2}{2} + x \right\}}$$

Set

$$\frac{d}{dx} \left\{ -\frac{x^2}{2} + x \right\} = -x + 1 = 0$$

$$\Rightarrow x = 1$$



So $x = 1$ is the maximizer

$$C = \int \frac{2}{\pi} e^{-\frac{1}{2} + 1} = \int \frac{2}{\pi} e^{\frac{1}{2}} = \sqrt{\frac{2e}{\pi}}$$

A - C:

① Generate $Y \sim \frac{1}{2} e^{-|y|}$, $-\infty < y < \infty$

② "Accept" Y with prob

$$\frac{f(Y)}{Cg(Y)} = \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{Y^2}{2}}}{\int \frac{2e}{\pi} \cdot \frac{1}{2} e^{-|Y|}} \\ = e^{-\frac{Y^2}{2} + |Y| - \frac{1}{2}}$$

otherwise "reject" and go back to ①

pseudo-code:

① Generate $Y \sim \frac{1}{2} e^{-|Y|}, -\infty < Y < \infty$

② Generate $U \sim \text{unif}(0, 1)$

③ If $U \leq e^{-\frac{Y^2}{2} + |Y| - \frac{1}{2}}$, set

$X = Y$; otherwise go back to ①

Output X .

To generate Y , we use ITM or
the following:

- Generate $W \sim \text{Exp}(1)$

(ie. $W = -\log(1 - V)$

where $V \sim \text{Unif}(0, 1)$)

- Generate $Z = \begin{cases} +1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$

(ie. Z is a "sign" variable)

Output Z.W

This is related to the composition
method

Implementation

In the first example, suppose the sequence of $\text{Unif}(0,1)$ generated by the computer is:

0.10 0.32 0.50 0.49 0.23 0.37 0.58 0.32 0.39 0.99

$$Y = 0.1$$

$$U = 0.32$$

Since $U > Y^2$, continue

$$Y = 0.5$$

$$U = 0.49$$

Since $U > Y^2$, continue

$$Y = 0.23$$

$$U = 0.37$$

Since $u > \gamma^2$, continue

$$\gamma = 0.58$$

$$u = 0.72$$

Since $u \leq \gamma^2$,

output 0.58

Implementation

In the first example, suppose the sequence of $\text{Unif}(0,1)$ generated by the computer is:

0.10 0.32 0.50 0.49 0.23 0.37 0.58 0.32 0.39 0.99

The A-R algorithm runs as:

$Y=0.10$

$U=0.32$

Since $U = 0.32 > Y^2 = 0.01$, continue

$Y=0.50$

$U=0.49$

Since $U = 0.49 > Y^2 = 0.25$, continue

$Y=0.23$

$U=0.37$

Since $U = 0.37 > Y^2 = 0.0529$, continue

$Y=0.58$

$U=0.32$

Since $U = 0.32 \leq Y^2 = 0.3364$, stop

Output = 0.58

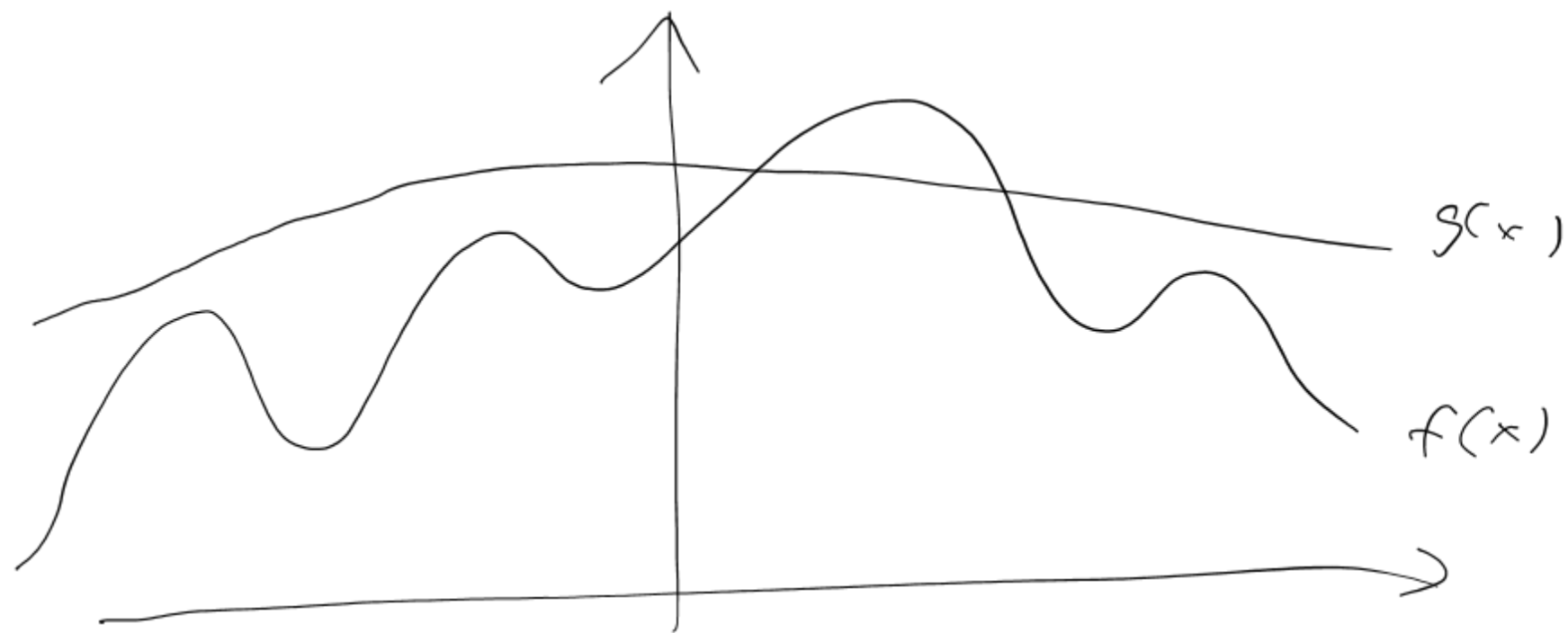
Intuition

Motivation:

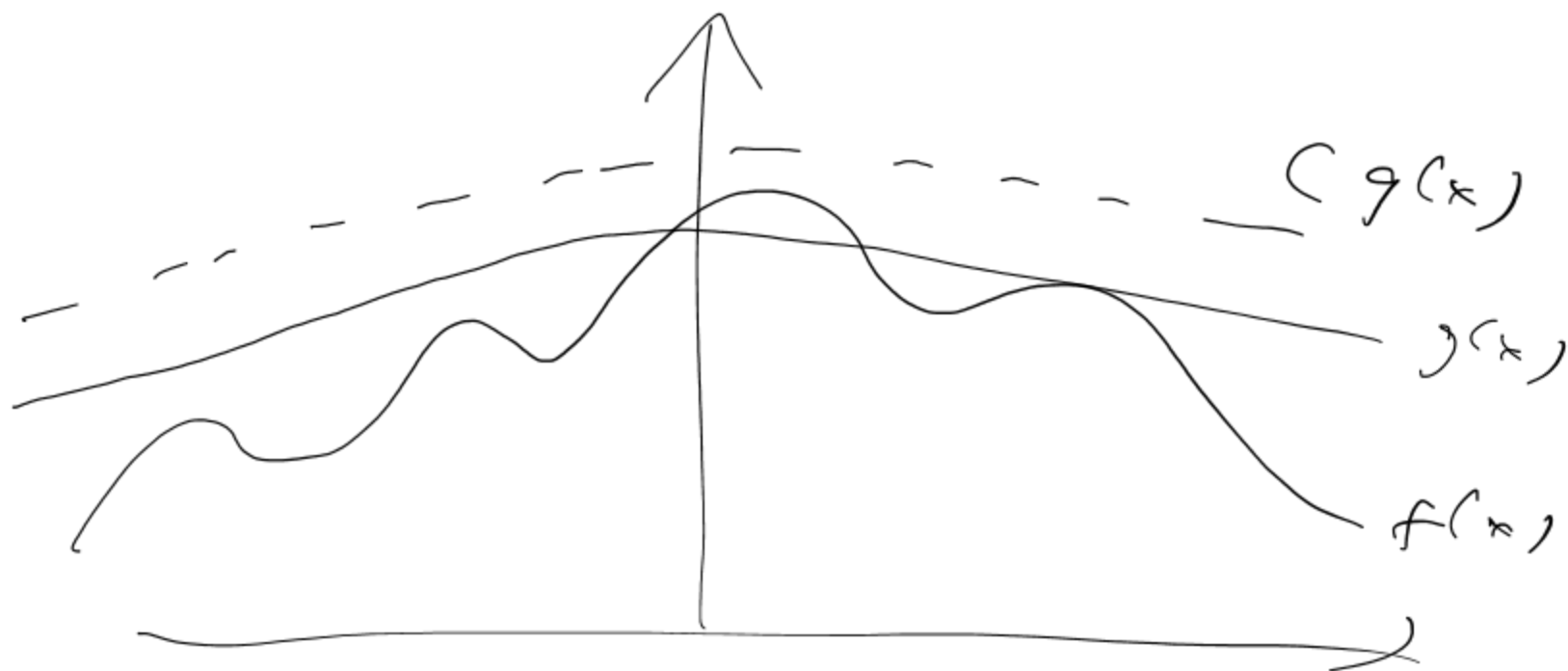
- Since $f(x)$ is difficult to sample from, we want to use an easier density $g(x)$
- However, we cannot just change the problem to sampling from $g(x)$

Mechanism:

- $Cg(x)$ is an “envelope” that covers $f(x)$
- C must be at least 1; if $C = 1$, then $f \equiv g$
- The acceptance probability $f(x)/Cg(x)$ is an adjustment from sampling $g(x)$ to sampling $f(x)$



g cannot be completely above f , because
if this is the case then one of them does not integrate to 1.



C is a constant that lifts g up so that $Cg(x)$ is always above $f(x)$.

In the procedure, we choose C such that

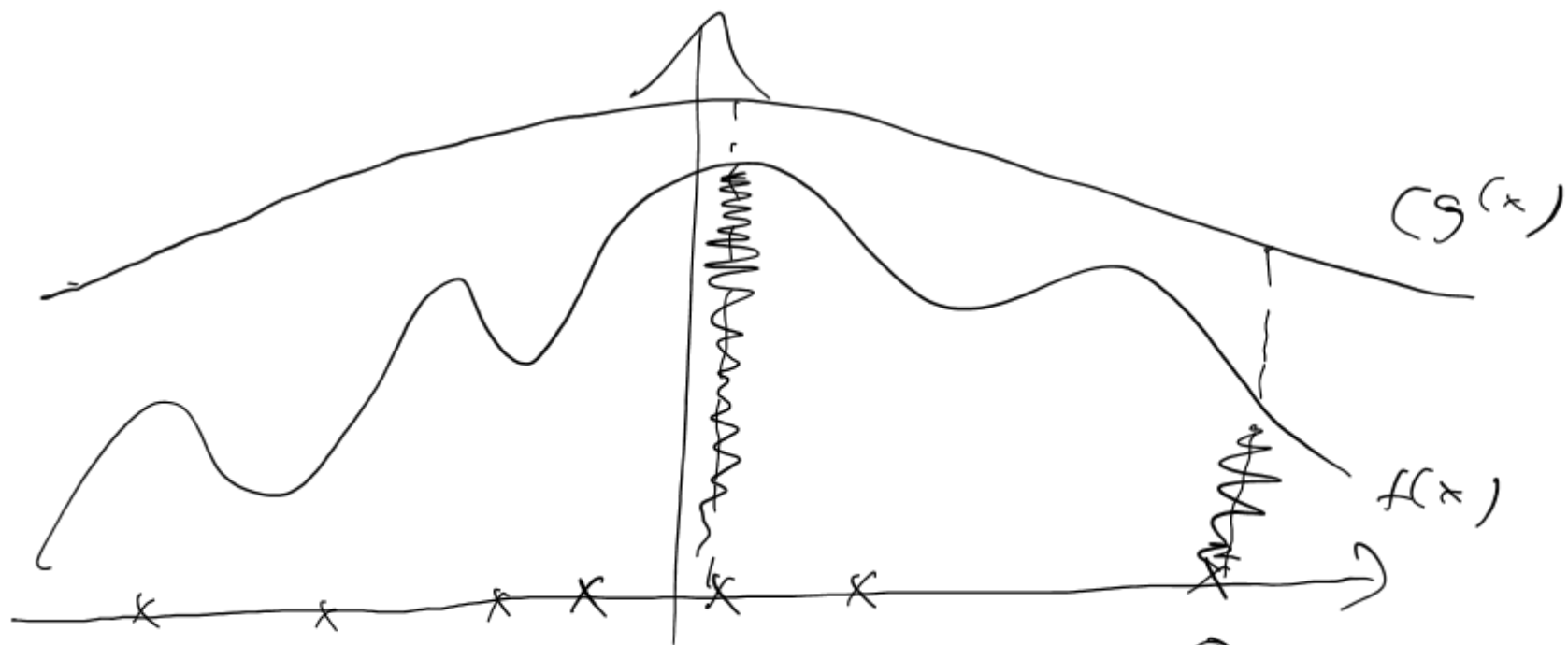
$$\frac{f(x)}{g(x)} \leq C \quad \text{for all possible } x.$$

$$\Leftrightarrow f(x) \leq C g(x) \quad \text{for all possible } x.$$

Note that $C \geq 1$

If we can choose $C=1$, this means $f=g$.

$\Rightarrow$ the A-C procedure is trivial.



"accept" sample from g with prob
that is $\frac{f(x)}{cg(x)}$.

sample from g .

The acceptance-rejection step is a mechanism to adjust sample from g into following f .

If f & cg are close at x , i.e.

$\frac{f(x)}{cg(x)} \approx 1$, accept with high prob,

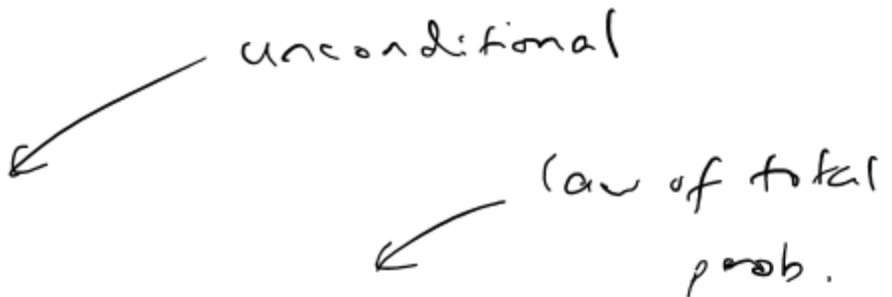
If f & cg are far apart at x , i.e.

$\frac{f(x)}{cg(x)} \approx 0$, accept with low prob.

Validity

The A-R algorithm indeed generates a random variable with density $f(x)$

First note that


$$\begin{aligned} P(\text{accept}) &= \int_{-\infty}^{\infty} P(\text{accept} | Y = x) g(x) dx \\ &= \int_{-\infty}^{\infty} \frac{f(x)}{C g(x)} \times g(x) dx \\ &= \frac{\int_{-\infty}^{\infty} f(x) dx}{C} \\ &= \frac{1}{C} \end{aligned}$$

where we use the law of total probability in the first equality

Validity

Next consider

$$\begin{aligned} & P(X \in (x, x + \Delta x) | \text{accept}) \\ &= \frac{P(X \in (x, x + \Delta x), \text{accept})}{P(\text{accept})} \\ &= \frac{P(Y \in (x, x + \Delta x)) \times P(\text{accept} | Y \in (x, x + \Delta x))}{P(\text{accept})} \\ &= \frac{g(x)\Delta x \times \frac{f(x)}{Cg(x)}}{\frac{1}{C}} \\ &= f(x)\Delta x \end{aligned}$$

- A-R is a valid algo, ie. it correctly outputs a r.v. $\sim f(x)$.
- A more efficient algo uses less runtime.
- A-R has a random runtime (# iterations to get 1 acceptance). A measurement in comparing algo's is the expected runtime.

runtime = # iterations to get 1 acceptance

$$\sim \text{Geo}(p) \quad , \quad p = P(\text{accept}) \\ = \frac{1}{C}$$

$$\text{expected runtime} = \frac{1}{p} = C$$

When C is large, expected runtime is large $\Rightarrow$ bad.

The smaller the C , the better.

If $C=1$, expected runtime $= 1 \Rightarrow$ always accept in 1st iteration.

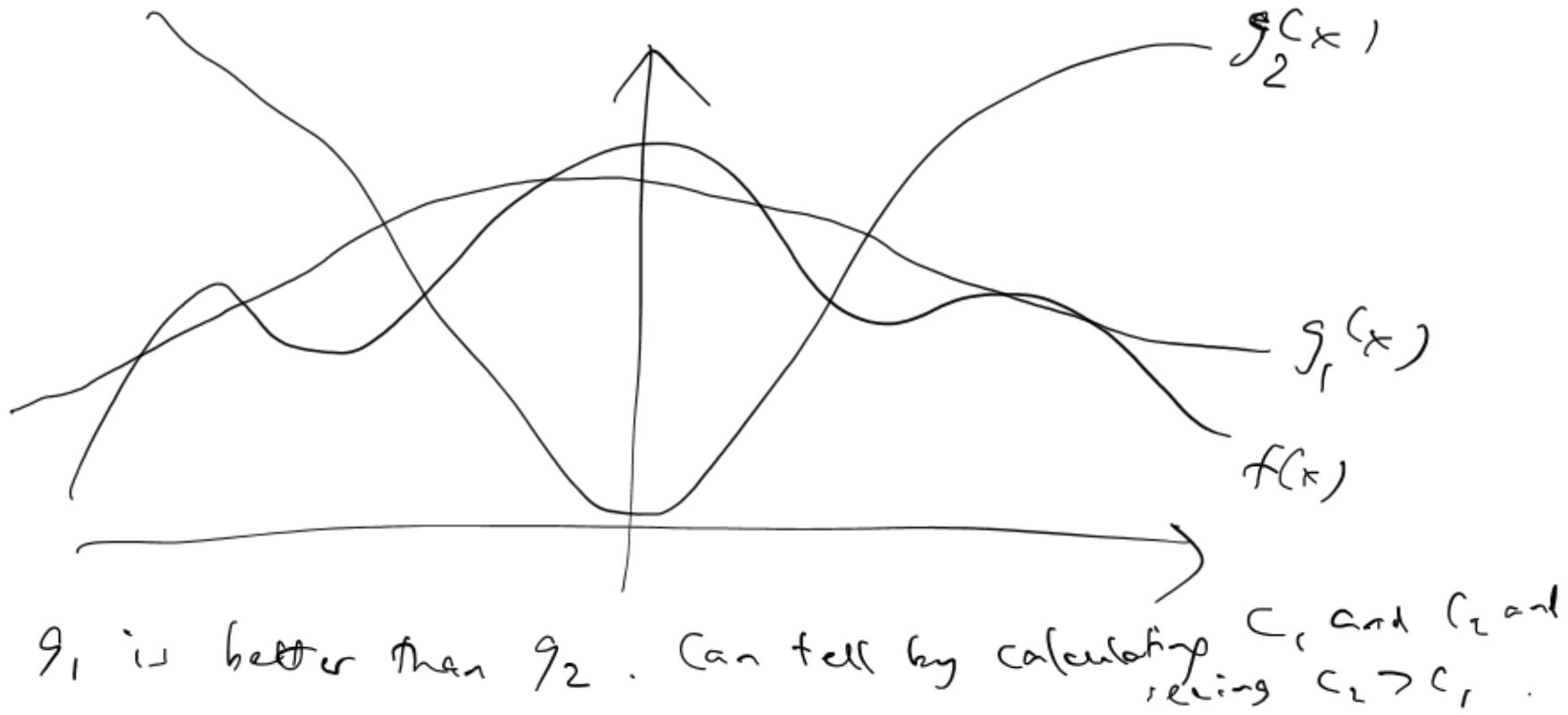
This is because $f = g$.

Efficiency

- The runtime of the algorithm depends on the number of trials to obtain one acceptance
- Since the algorithm requires a random number of trials, it makes sense to consider the expected number of trials as an efficiency measurement.
- The number of trials to get one acceptance is $\text{Geo}(p)$, where p is the acceptance probability
- The acceptance probability is $p=1/C$, where C is the constant that bounds $f(x)/g(x)$
- In other words, the expected number of trials to get one acceptance is $1/p=C$

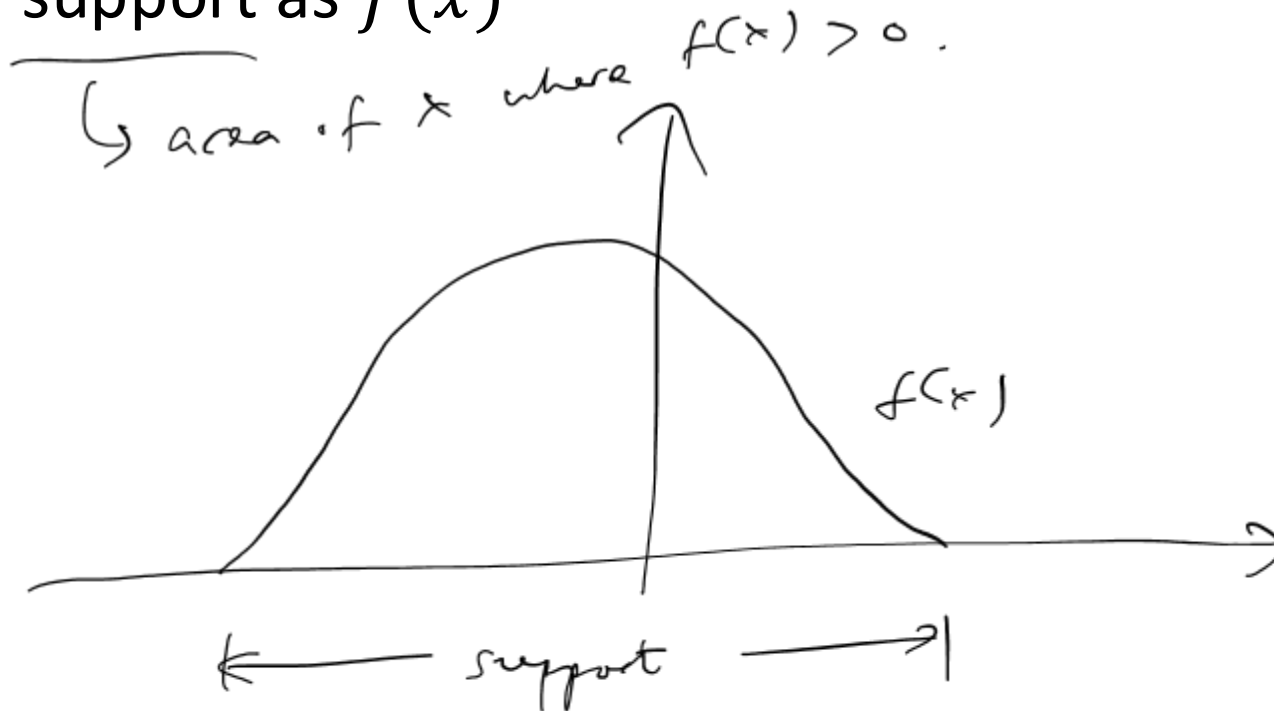
Efficiency

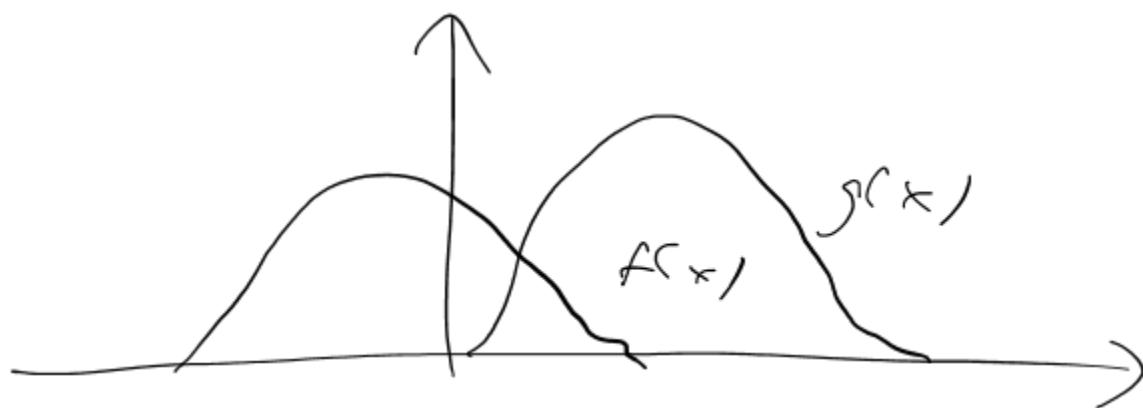
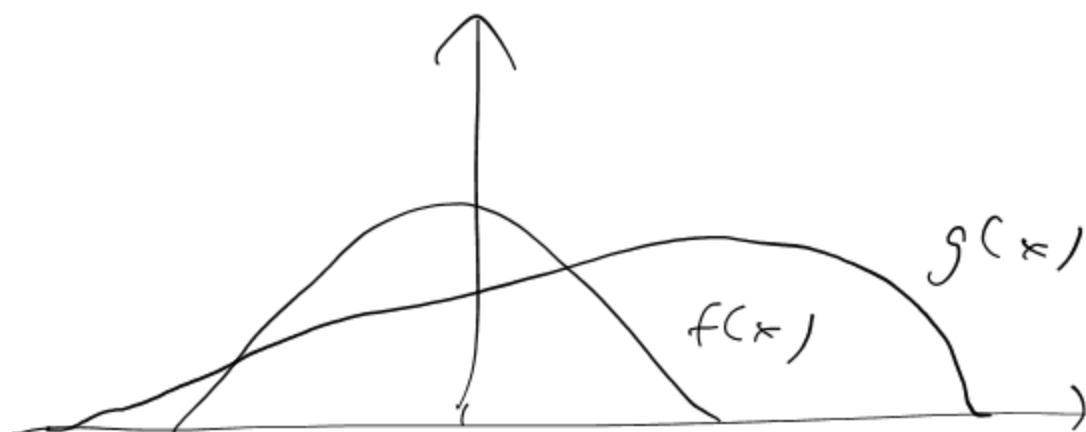
- If $g(x)$ is close to $f(x)$, then C is small, so p is large, and expected number of trials is small
- If $g \equiv f$, then $C=1$, so that $p=1$ and we finish the random variable generation always in one trial



How to Choose Proposal

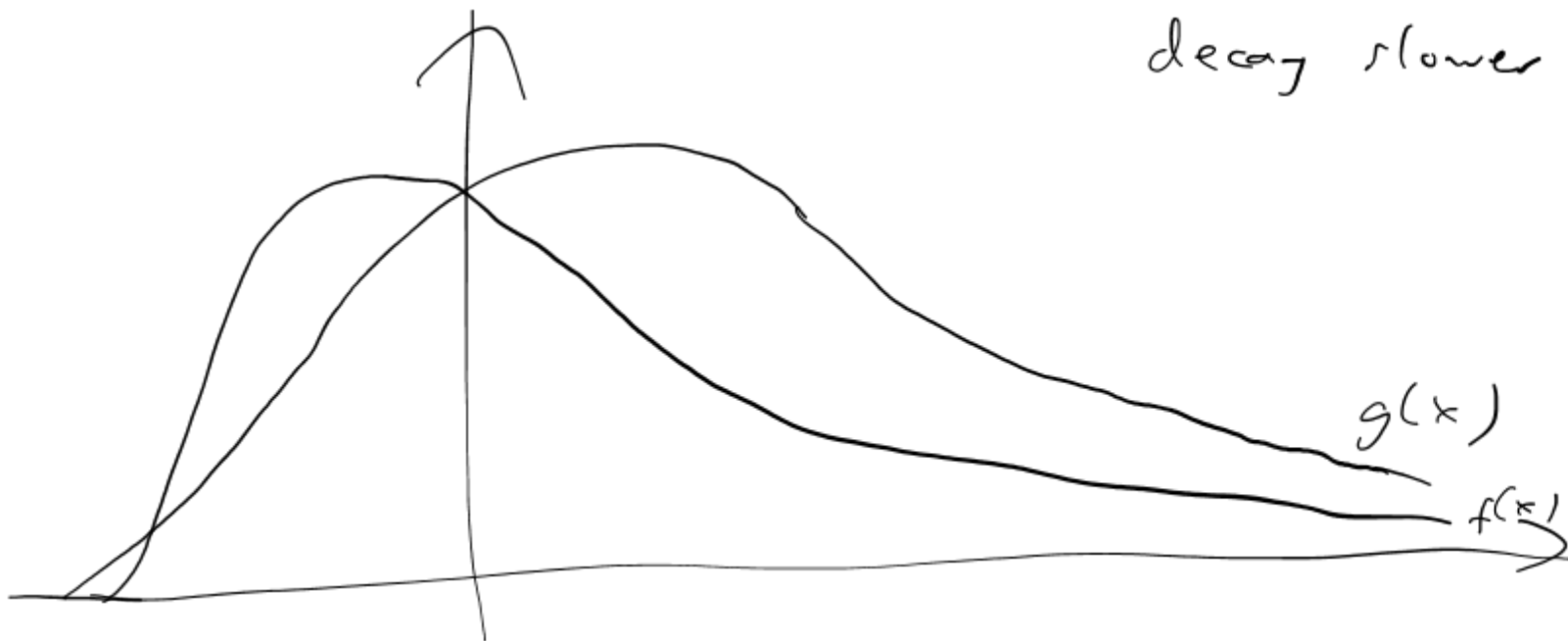
- Before we even consider efficiency, we have to make sure the basic requirement of $g(x)$ holds
- Requirement 1: $g(x)$ should have at least the same support as $f(x)$





How to Choose Proposal

- Requirement 2: $g(x)$ should have a heavier tail than $f(x)$ in the case that $f(x)$ has unbounded support



How to Choose Proposal

Most generally, we must be able to find C that is finite; otherwise, the A-R procedure fails

How to Choose Proposal

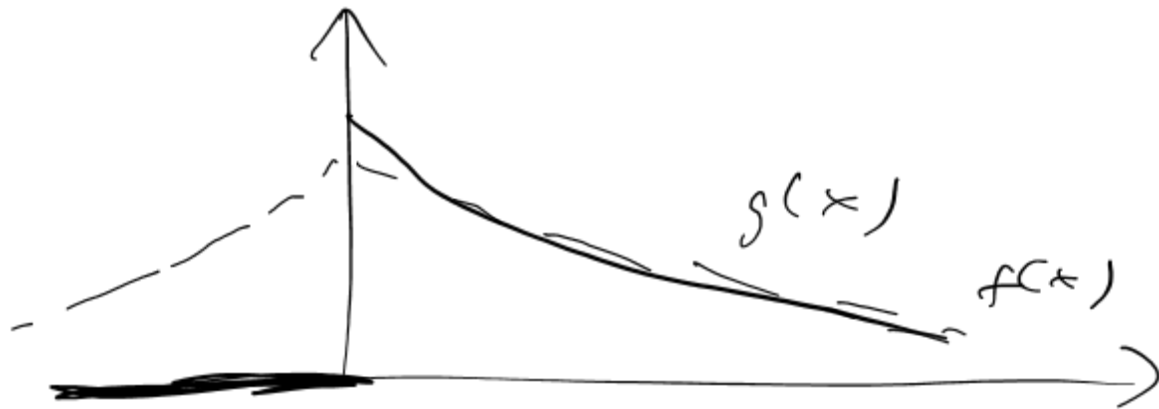
Can we use the following to generate $\text{Exp}(1)$, i.e., with density $f(x) = e^{-x}$ for $x > 0$ and 0 otherwise:

- Two-sided exponential, i.e., $g(x) = \frac{1}{2} e^{-|x|}$, $-\infty < x < \infty$
- $\text{Unif}(0,1)$, i.e., $g(x) = 1$ for $0 < x < 1$ and 0 otherwise

Can we use the following to generate $N(0,1)$, i.e., with density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty:$$

- $\text{Exp}(1)$, i.e., $g(x) = e^{-x}$ for $x > 0$ and 0 otherwise
- Density $g(x) = K e^{-|x|^3}$ for some $K > 0$

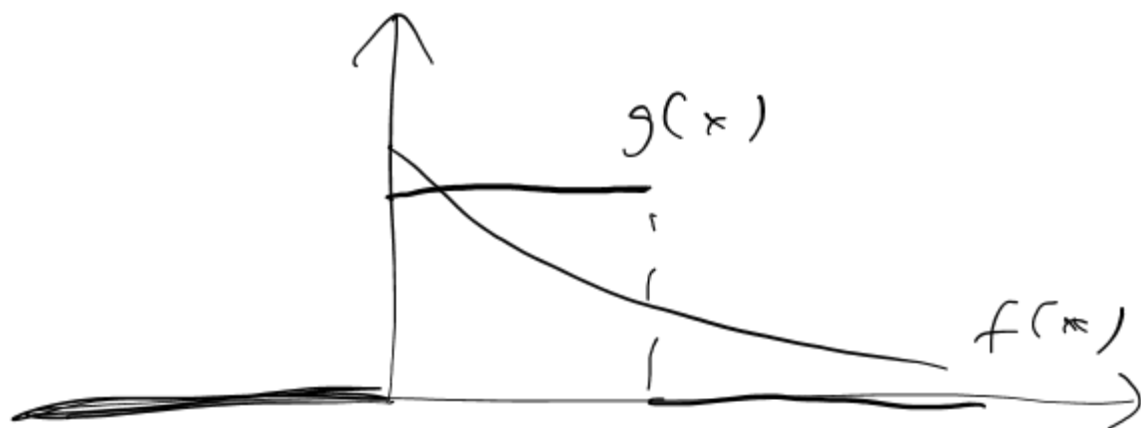


Find C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for } x > 0$$

$$\Rightarrow \frac{e^{-x}}{\frac{1}{2}e^{-|x|}} = 2 \quad \Rightarrow \text{we can choose } C=2$$

This also means Requirements 1 and 2 hold.
and the A-R algo is valid.



Find C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for } x > 0.$$

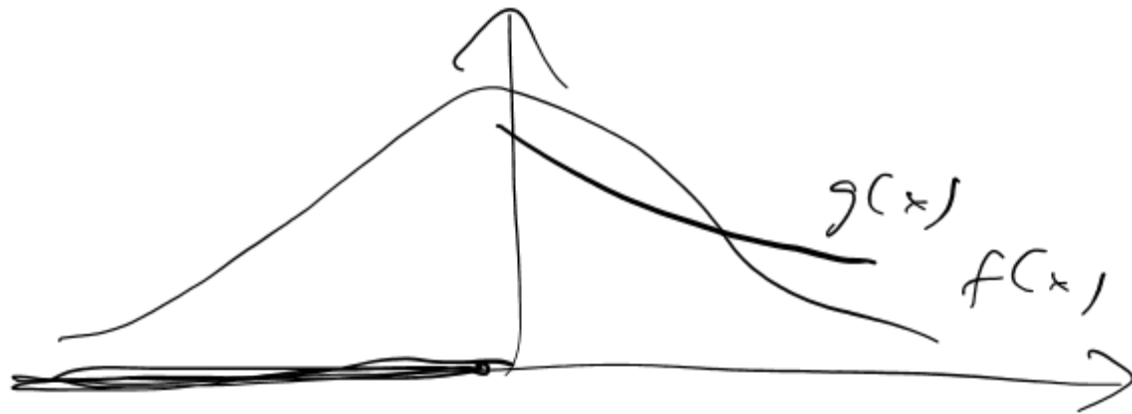
$$\Rightarrow \frac{f(x)}{g(x)} = \frac{e^{-x}}{1} \quad \text{for } 0 < x < 1 \quad \text{and} \quad \frac{e^{-x}}{0} \quad \text{for } x > 1$$

There is no finite C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for } x > 0.$$

The algo is invalid.

We can see that Requirement 1 is violated.



Requirement 1 is violated because
when $x < 0$, $f(x) > 0$ but $g(x) = 0$.
 $\Rightarrow$ Exp(1) is invalid proposal.

find C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for } -\infty < x < \infty$$

$$\frac{f(x)}{g(x)} = \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}{k e^{-|x|^3}} = \frac{1}{\sqrt{2\pi} k} e^{-\frac{x^2}{2} + |x|^3}$$

Since $-\frac{x^2}{2} + |x|^3 \rightarrow \infty$ as $x \rightarrow \pm\infty$

so $\frac{f(x)}{g(x)}$ is unbounded as $x \rightarrow \pm\infty$

We cannot find a finite C such that

$$\frac{f(x)}{g(x)} \leq C \quad \text{for} \quad -\infty < x < \infty.$$

$\Rightarrow g(x) = k e^{-|x|^3}$ is invalid proposal.